

MOHSEN MOSTAFA

Co-Founder & Chief AI Scientist, Aventro | AI Team Lead & Research Engineer

Independent Researcher — Physical & Geometric AI

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PROFILE

Co-founder and Chief AI Scientist building Aventro, an R&D-driven venture at the intersection of epistemology, world models, and geometric & physical AI applied to software robotics and computer vision. 8+ years in AI/ML/CV/GenAI and 10+ years in IT infrastructure, with a self-directed research program producing 8 first-author papers on geometric deep learning, Bayesian uncertainty-aware architectures, and physics-informed optimization — all validated on free-tier cloud GPUs. Combines hands-on production AI leadership (multimodal CV+LLM systems, MLOps at scale) with original theoretical contributions, now extending that methodology toward embodied prediction, world models, and ROS2-based robotics.

FOUNDER EXPERIENCE

Co-Founder & Chief AI Scientist — Aventro (aventro.co.uk) (2026 – Present)

- Co-founded Aventro as an R&D venture unifying epistemology, world models, and geometric & physical AI, applied to software robotics and computer vision.
- Set technical direction and research agenda: originating and publishing new methods that extend prior work in geometric deep learning, uncertainty-aware architectures, and physics-informed optimization toward embodied, world-model-driven systems.
- Designing the Data & AI infrastructure layer for the venture — data pipelines, training/serving infrastructure, and MLOps tooling — to support research-to-production for physical AI and robotics workloads.
- Building software robotics capability on ROS2, bridging research-grade perception and world-model components with real robot middleware and control stacks.
- Owns technical narrative and roadmap for investor, partner, and ecosystem conversations (incl. accelerator/ecosystem applications such as Hub71).

PROFESSIONAL EXPERIENCE

AI Lead & Computer Vision Researcher — GraviLog (Remote — US/UAE) (Sep 2025 – Present)

- Designed end-to-end multimodal AI system architecture (CV + LLM) with TensorRT/ONNX optimizations, achieving a 40% latency reduction.
- Defined the technical roadmap for 3 products and delivered strategic presentations to C-level stakeholders.
- Led the RAG retrieval pipeline (Pinecone + embeddings); mentored 4 engineers in MLOps and system design.

Senior AI Developer (Research Focus) — OMC Technologies (Remote — Limerick, Ireland) (May 2023 – Aug 2025)

- Built AI agents integrating LLMs (Falcon, DeepSeek, Qwen, Llama), automating 3 workflows and cutting manual effort by 40%.
- Deployed a production RAG system (LangChain + Pinecone) that reduced hallucinations by 35%.
- Shipped a multimodal healthcare AI system: F1 +19%, latency reduced from 210ms to 145ms via streaming + FP16.
- Built a fraud detection system: AUC +17%, false positives -22%; FastAPI service at p99 ≤ 95ms @ 500 RPS.
- Architected an InsurTech PoC for automated roof damage assessment, fine-tuning SAM 3 via Roboflow on drone imagery, with a custom mIoU-to-area and MAPE-driven cost estimation pipeline.
- Managed AI infrastructure spanning 24 GPUs (12 nodes) and 15 CPU nodes for real-time IoT anomaly detection; optimized distributed training/inference with TensorRT, Kubeflow, and KServe.
- Mentored 5 junior engineers, leading to 2 promotions.

AI Developer & Researcher — Nexgro (South Africa) (Apr 2020 – Apr 2023)

- Built a grain quality computer vision pipeline achieving 98% accuracy across 6 grain types.

- Delivered drone/satellite anomaly detection (NDVI + YOLOv8), raising manual inspection efficiency by 160%.
- Optimized 3D point cloud segmentation for LiDAR/DSM data, reaching sub-300ms inference on edge devices.
- Deployed an LLM RAG assistant over technical manuals, saving 15 person-hours per week.

Data Scientist — Nexgro (South Africa) *(Mar 2018 – Mar 2020)*

- Built forecast models that reduced logistics costs by 18%.
- Predictive maintenance models prevented 3 breakdowns, saving roughly \$45K.
- Inventory optimization reduced spoilage by 18%.

Systems Administrator — Egyptian Co. *(Jun 2013 – Mar 2018)*

- Managed IT infrastructure for 200+ users, sustaining 99.8% uptime across multi-site networks and servers.
- Automation scripts cut manual ticket handling by 25%; self-directed pivot into data science (certifications 2016–2018).

IT Network Administrator / Specialist — International Co. for Trade *(Aug 2005 – Jun 2013)*

- Administered LAN/WAN, firewalls, and backup solutions for 150+ nodes; first-line network support and vendor coordination.

INDEPENDENT RESEARCH — PHYSICAL & GEOMETRIC AI

Self-directed research program (2023–present) defining "Physical & Geometric AI": geometric foundations of representation learning, Bayesian/uncertainty-aware architectures, and physics-informed, curvature-aware optimization — 8 first-author preprints completed in under 10 weeks of active writing, validated entirely on free-tier cloud GPUs (Kaggle/Colab T4 15GB, P100 16GB). Currently extending this methodology toward embodied prediction and world models (LePhyJEPA; Graph-Grounded Safe Routing for Adaptive World Models).

- Geometric Foundations of Racing Dynamics: How Gradient Descent Adapts Network Capacity on Data Manifolds (2026) — Preprints.org 202603.0760/v1
- Gaussian-Constrained LeJEPA Representations for Unsupervised Scene Discovery (2025) — arXiv:2602.07016
- HPST: A Hybrid Physics-Spectral-Threshold Framework for Fluid Flow Analysis (2026) — Preprints.org 202603.1281/v1
- R-LayerNorm: Robust Layer Normalization with Adaptive Noise Suppression (2025) — Preprints.org 202603.0060
- Bayesian R-LayerNorm: Uncertainty-Aware Adaptive Normalization with Provable Robustness Bounds (2026) — Preprints.org 202603.0740/v1
- Bayesian PASA: Probabilistic Adaptive Sigmoidal Activation with Uncertainty Quantification (2026) — Preprints.org
- Uncertainty-Aware Classifier with Physics-Based Rejection (UA-PBR) (2026) — Preprints.org 202603.0748/v1
- EPANG-Gen: A Curvature-Aware Optimizer with Uncertainty Quantification for Scientific ML (2026) — Preprints.org 202603.0692/v1
- LePhyJEPA: Integrating Geometric Priors into Latent Predictive Architectures for Multi-Modal Representation Learning — in progress

Full list on ORCID: 0009-0004-4478-0317

EDUCATION

B.Sc. Management Information Systems — High Institute for Specialized Technological Studies, Cairo, Egypt (2000 – 2005)

TECHNICAL SKILLS

- Deep Learning: PyTorch, TensorFlow, Hugging Face (Falcon, Llama, Mistral, Qwen), JEPA, CLIP, YOLO, Mask2Former
- Probabilistic ML: Bayesian deep learning, uncertainty quantification, natural gradient methods, curvature-aware optimization
- GenAI / LLM: RAG, ColBERT, LangChain, DeepSeek, RLHF concepts, streaming inference
- Computer Vision & Robotics: OpenCV, 3D point cloud segmentation (LiDAR/DSM), multi-view geometry, ROS2, software robotics
- MLOps & DevOps: MLflow, Kubeflow, Airflow, Docker, Kubernetes, KServe, FastAPI, GitLab CI, GCP Vertex AI, AWS SageMaker
- Streaming & Big Data: Apache Kafka, Spark Structured Streaming, Hadoop HDFS, Parquet
- Optimization & Serving: TensorRT, ONNX, model pruning/quantization, FP16 inference
- Data & Databases: Pandas, Dask, Polars, PySpark, SQL (PostgreSQL), MongoDB, Redis, Pinecone

CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI
- TensorFlow Developer & Advanced TensorFlow — Google
- Google Cloud MLOps — Google
- Total Data Quality — University of Michigan
- Research Methodology — University of London
- IBM Data Science Professional Certificate — IBM
- CCNA 200-301 & ENCORE 350-401 — Cisco

LEADERSHIP & LANGUAGES

- Mentored 5+ engineers across OMC Technologies and GraviLog, resulting in 2 direct promotions.
- Conducted internal LLM/MLOps workshops and knowledge-transfer sessions.
- Designed a full MLOps showcase pipeline: Kafka → Spark Streaming → HDFS → MongoDB → PostgreSQL → MLflow → Kubeflow → FastAPI for real-time predictive maintenance.
- Languages: Arabic (native), English (fluent, C1/C2).